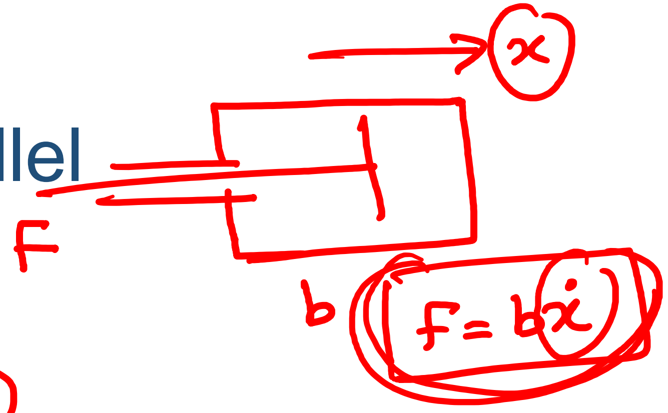
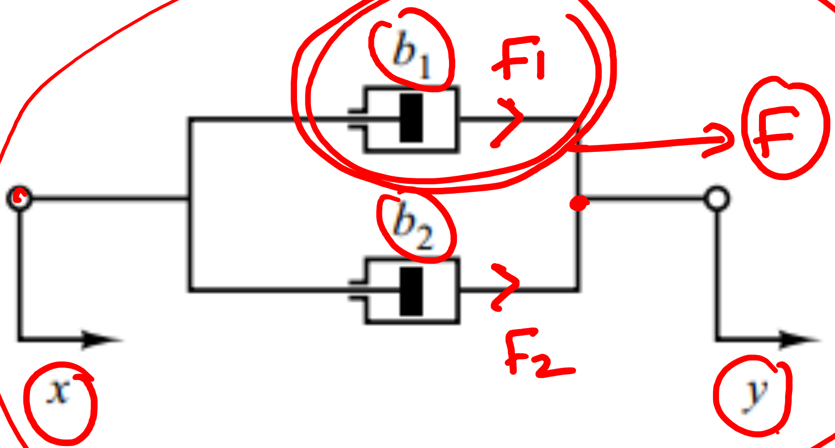
Mechanical System Modelling

System consisting of two dampers connected in parallel



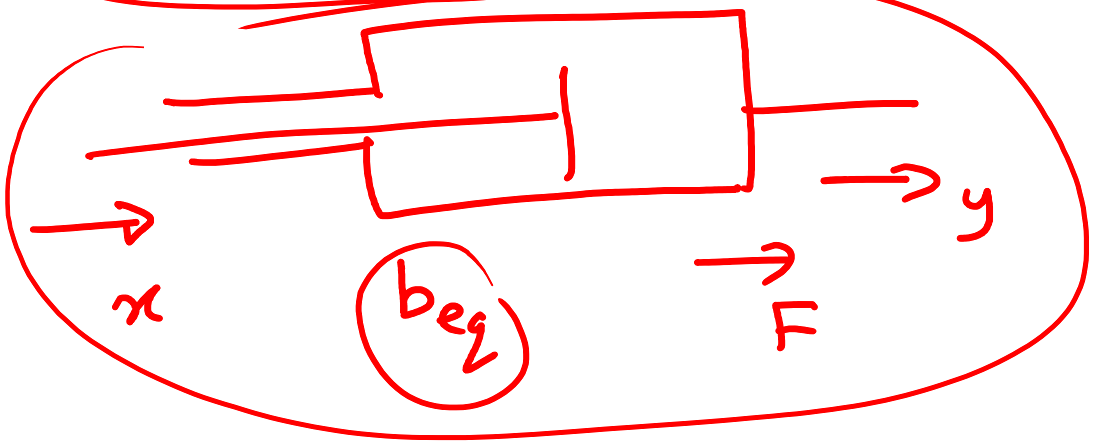


$$F = F_1 + F_2$$

$$F = b_1(\dot{y} - \dot{x}) + b_2(\dot{y} - \dot{x})$$

$$F = (b_1 + b_2)(\dot{y} - \dot{x}) \quad \text{--- } ①$$

$$F = b_{eq}(\dot{y} - \dot{x}) \quad \text{--- } ②$$

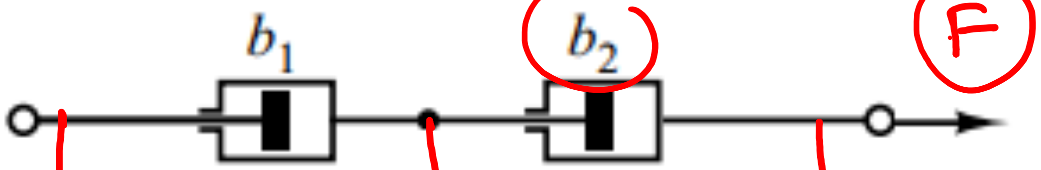


① = ② $\Rightarrow$

$$b_{eq} = b_1 + b_2$$

Exercise: Mechanical System Modelling

System consisting of two dampers connected in series



F

x $\quad$ z $\quad$ y

$(z - x)$ $\quad$ $(y - z)$

$(\dot{z} - \dot{x})$ $\quad$ $(\dot{y} - \dot{z})$

b_{eq} ?

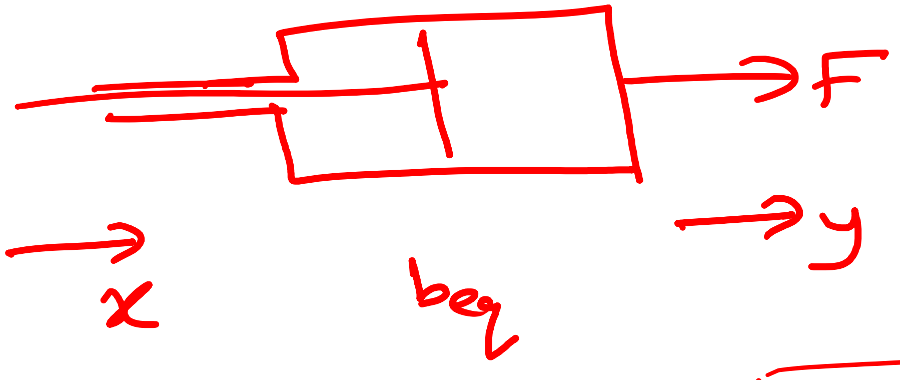
$F = b\dot{x}$

$F = b_1(\dot{z} - \dot{x}) = b_2(\dot{y} - \dot{z})$

$\Downarrow$

$\dot{z} =$

$F(\dot{x}, \dot{y}) \Rightarrow$



$$F = b_{eq}(\dot{y} - \dot{x})$$

$$\boxed{b_{eq} = \frac{b_1 b_2}{b_1 + b_2}}$$

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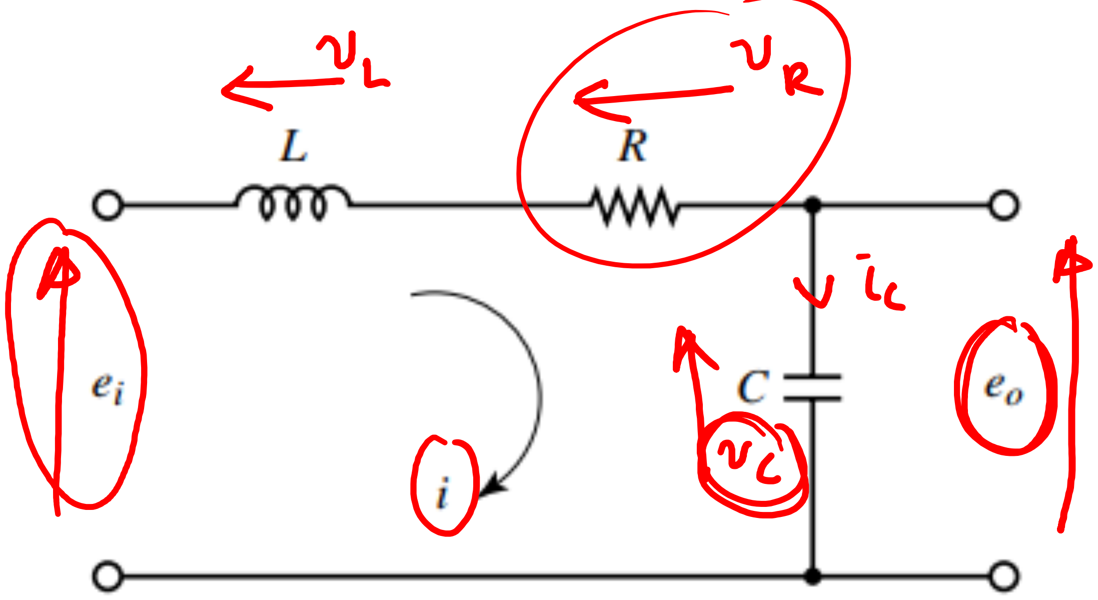
Electrical System Modelling

Consider the electrical circuit shown in the Figure



$$v_L = L\frac{di}{dt}$$

$$v_R = Ri$$

$$i_c = C\frac{dv_c}{dt}$$

$$\Downarrow$$

$$v_c = \frac{1}{C}\int i\,dt$$

$$\boxed{e_i = v_L + v_R + v_c = L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt}$$

$$v_c = e_i - v_L - v_R$$

Transfer Functions

- Describe the relationship between O/P and I/P
- Transform it into the Laplace domain

$$\boxed{\dot{x}(t) + \frac{1}{\tau}x(t) = Ku(t)}$$

$$sx(s) + \frac{1}{\tau}x(s) = K\,u(s)$$

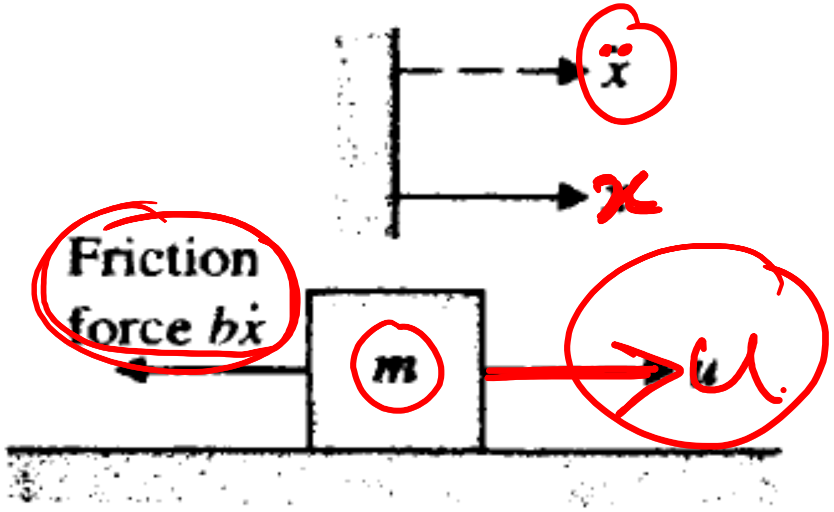
$$x(s)\left\{s + 1/\tau\right\} = K\,u(s)$$

$$\boxed{\frac{x(s)}{u(s)} = \frac{K}{s + 1/\tau}}$$

Exercise: Transfer Functions

$$v = \dot{x}$$

$$F = ma$$



$$u - b\dot{x} = m\ddot{x}$$

$$u = m\ddot{x} + b\dot{x}$$

$$\mathcal{L} \Rightarrow U(s) = ms^2 X(s) + bsX(s)$$

$$U(s) = X(s)\{ms^2 + bs\}$$

$$\boxed{\frac{X(s)}{U(s)} = \frac{1}{ms^2 + bs}}$$

$$V(s) = sX(s)$$

$$X(s) = V(s)/s$$

$$\frac{V(s)/s}{U(s)} = \frac{1}{\cancel{s}(b + ms)}$$

$$\boxed{\frac{V(s)}{U(s)} = \frac{1}{ms + b}}$$